ELE2038 Control Coursework Report

Modelling and Control of a Wooden Ball on an Inclined Plane

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Abstract

This report presents the modelling, linearisation, and control-oriented analysis of a nonlinear electromechanical system consisting of a wooden ball rolling on an inclined plane under spring-damper forces and electromagnetic actuation. A nonlinear state-space model is derived from first principles using mechanical and electrical laws. The model is then linearised about an operating point near $x_{sp} = 0.5\,\mathrm{m}$. By applying the Laplace transform to the linearised equations, a transfer function is obtained and used to study the open-loop dynamics. The open-loop system is shown to be unstable, which motivates the introduction of feedback control. A PID controller is selected because it provides a practical balance between fast response, zero steady-state error, and damping. The report also discusses disturbance rejection, robustness, practical limitations, and the structure of the closed-loop system.

1 Introduction

The aim of this coursework is to design a controller for a nonlinear system so that the ball position is regulated near a desired set-point. The closed-loop system should satisfy the following objectives:

- BIBO stability,
- zero steady-state error,
- limited oscillatory behaviour,
- rejection of realistic disturbances.

The original plant is nonlinear because both the magnetic force and the inductance depend on the ball position. For this reason, the modelling process begins with a nonlinear description of the system. A local linear model is then obtained around a chosen equilibrium point and used for transfer-function analysis and controller design.

2 System Description

The system consists of a wooden ball of mass m and radius r rolling on an inclined plane at angle ϕ. The ball is connected to a spring of stiffness k and a damper with viscous coefficient b. An electromagnet located at position $x = \delta$ attracts a small metal core inside the ball. The electrical input is the applied voltage V, and the mechanical output of interest is the position of the ball along the incline.

3 Nonlinear Modelling

3.1 State Definition

For the simplified model used in the final derivation, the state-space representation is written using the mechanical and electrical states only. The sensor dynamics are introduced later when discussing the measured output and the controller structure. The states are defined as

$$x_1 = x, \qquad x_2 = \dot{x}, \qquad x_3 = i.$$

The system input is

$$u = V,$$

and the output for this simplified model is taken as

$$y = x.$$

3.2 Mechanical Dynamics

Let x denote the position of the centre of the ball along the inclined plane. The distance between the ball and the electromagnet is

$$y_m = \delta - x.$$

The magnetic force magnitude is

$$F_{\mathrm{mag}} = \frac{c\,i^2}{(\delta - x)^2}.$$

Applying Newton's second law along the incline and accounting for rolling without slipping gives

$$\ddot{x} = \frac{5}{7m}\left(mg\sin\phi + \frac{c\,i^2}{(\delta - x)^2} - k(x - d) - b\dot{x}\right).$$

Therefore,

$$\dot{x}_1 = x_2$$

and

$$\dot{x}_2 = \frac{5}{7m}\left(mg\sin\phi + \frac{c\,x_3^2}{(\delta - x_1)^2} - k(x_1 - d) - bx_2\right).$$

3.3 Electrical Dynamics

The inductance depends on the ball position according to

$$L(x) = L_0 + L_1 e^{-\alpha(\delta - x)}.$$

Using the simplified electrical model,

$$V = iR + L(x)\dot{i},$$

so

$$\dot{i} = \frac{V - Ri}{L_0 + L_1 e^{-\alpha(\delta - x)}}.$$

Hence,

$$\dot{x}_3 = \frac{u - Rx_3}{L_0 + L_1 e^{-\alpha(\delta - x_1)}}.$$

3.4 Final Nonlinear State-Space Model

The nonlinear state-space model is

$$\dot{x}_1 = x_2, \tag{1}$$

$$\dot{x}_2 = \frac{5}{7m}\left(mg\sin\phi + \frac{c\,x_3^2}{(\delta - x_1)^2} - k(x_1 - d) - bx_2\right), \tag{2}$$

$$\dot{x}_3 = \frac{u - Rx_3}{L_0 + L_1 e^{-\alpha(\delta - x_1)}}. \tag{3}$$

3.5 Assumptions

The main assumptions used in the model are:

- the ball rolls without slipping,
- the ball is approximated as a solid sphere,
- spring and damper forces act along the incline,
- the electrical circuit is modelled using a simplified inductor-resistor relation,
- the sensor dynamics are introduced later in the controller design section.

3.6 Modelling Discussion

The model is nonlinear because the magnetic force depends on x_3^2 and $(\delta - x_1)^{-2}$, while the electrical dynamics also depend on position through the inductance term. These nonlinearities make direct controller design difficult and motivate the use of local linearisation around an equilibrium point.

4 Equilibrium Point

An equilibrium point $(x_1^e, x_2^e, x_3^e, u^e)$ satisfies

$$0 = x_2^e,$$

$$0 = \frac{5}{7m}\left(mg\sin\phi + \frac{c(x_3^e)^2}{(\delta - x_1^e)^2} - k(x_1^e - d) - bx_2^e\right),$$

$$0 = \frac{u^e - x_3^e R}{L_0 + L_1 e^{-\alpha(\delta - x_1^e)}}.$$

Choosing the operating point

$$x_1^e = x_{sp} = 0.5, \qquad x_2^e = 0,$$

the equilibrium current is obtained from

$$0 = mg\sin\phi + \frac{c(i_{sp})^2}{(\delta - x_{sp})^2} - k(x_{sp} - d),$$

which gives

$$\frac{c(i_{sp})^2}{(\delta - x_{sp})^2} = k(x_{sp} - d) - mg\sin\phi$$

and therefore

$$i_{sp} = \sqrt{\frac{1}{c}(\delta - x_{sp})^2\,[k(x_{sp} - d) - mg\sin\phi]}.$$

Numerically,

$$i_{sp} \approx 0.6988 \text{ A}.$$

From the electrical equilibrium equation,

$$u^e = V_{sp} = Ri_{sp},$$

so

$$V_{sp} \approx 1537.39 \text{ V}.$$

4.1 Practical Interpretation

The equilibrium voltage is very high. This indicates that the simplified model leads to a mathematically valid operating point, but one that may not be practically achievable without considering actuator limitations or alternative parameter choices. This observation is important when discussing practical feasibility in the report and in the interview.

5 Linearisation

5.1 Deviation Variables

Define the deviation variables around the equilibrium point:

$$\bar{x}_1 = x_1 - x_1^e, \qquad \bar{x}_2 = x_2 - x_2^e, \qquad \bar{x}_3 = x_3 - x_3^e, \qquad \bar{u} = u - u^e.$$

5.2 Linearised Equations

The linearised model is written as

$$\dot{\bar{x}}_1 = \bar{x}_2,$$

$$\dot{\bar{x}}_2 = A\bar{x}_1 + B\bar{x}_2 + C\bar{x}_3,$$

$$\dot{\bar{x}}_3 = D\bar{x}_1 + E\bar{x}_3 + F\bar{u},$$

where the coefficients are obtained from the partial derivatives of the nonlinear system evaluated at the equilibrium point:

$$A = \frac{5}{7m}\left(\frac{2c(x_3^e)^2}{(\delta - x_1^e)^3} - k\right),$$

$$B = -\frac{5b}{7m},$$

$$C = \frac{10}{7m}\left(\frac{cx_3^e}{(\delta - x_1^e)^2}\right),$$

$$D = -\frac{L_1\alpha(-Rx_3^e + u^e)e^{-\alpha(\delta - x_1^e)}}{\left(L_0 + L_1 e^{-\alpha(\delta - x_1^e)}\right)^2},$$

$$E = -\frac{R}{L_0 + L_1 e^{-\alpha(\delta - x_1^e)}},$$

$$F = \frac{1}{L_0 + L_1 e^{-\alpha(\delta - x_1^e)}}.$$

5.3 Accuracy of the Linearisation

The linearised system is only valid near the operating point. It is suitable for local stability analysis and controller design for small deviations around $x_{sp} = 0.5\,\mathrm{m}$. For larger changes in position or current, the original nonlinear effects become more significant and the linear approximation becomes less reliable.

6 Laplace Transform and Transfer Function

6.1 Laplace Domain Equations

Applying the Laplace transform to the linearised equations with zero initial conditions gives

$$s\bar{X}_1(s) = \bar{X}_2(s),$$

$$s\bar{X}_2(s) = A\bar{X}_1(s) + B\bar{X}_2(s) + C\bar{X}_3(s),$$

$$s\bar{X}_3(s) = D\bar{X}_1(s) + E\bar{X}_3(s) + F\bar{U}(s).$$

6.2 Transfer Function Derivation

From the first equation,

$$\bar{X}_2(s) = s\bar{X}_1(s).$$

Substituting into the second equation:

$$s^2\bar{X}_1(s) = A\bar{X}_1(s) + Bs\bar{X}_1(s) + C\bar{X}_3(s),$$

so

$$\bar{X}_3(s) = \frac{s^2 - A - Bs}{C}\bar{X}_1(s).$$

Substituting this into the third equation:

$$s\left(\frac{s^2 - A - Bs}{C}\right)\bar{X}_1(s) = D\bar{X}_1(s) + E\left(\frac{s^2 - A - Bs}{C}\right)\bar{X}_1(s) + F\bar{U}$$

Rearranging gives the transfer function from voltage deviation to position deviation:

$$\frac{\bar{X}_1(s)}{\bar{U}(s)} = \frac{CF}{s^3 + s^2(-B - E) + sBE - A - CD + AE}.$$

Therefore,

$$\boxed{G_s(s) = \frac{\bar{X}_1(s)}{\bar{U}(s)} = \frac{CF}{s^3 + s^2(-B - E) + sBE - A - CD + AE}}$$

7 Block Diagram Representation

At this stage, the linearised plant is incorporated into a closed-loop control structure. The purpose of this structure is to regulate the position of the ball around a desired set-point.

The reference input is denoted by $X^{sp}(s)$, which represents the desired position of the ball. The system output is $X(s)$, and the sensor produces a measured signal $G_m(s)X(s)$. The error signal is therefore defined as

$$E(s) = X^{sp}(s) - G_m(s)X(s).$$

The controller processes this error and generates the control input $U(s)$, which is the voltage applied to the system.

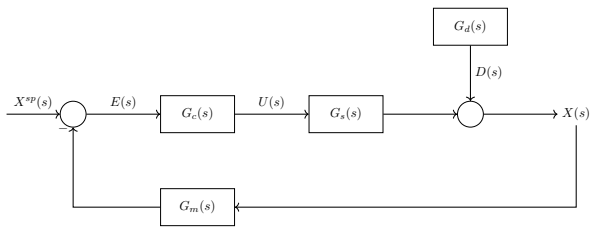

Figure 1: Closed-loop control structure including controller, plant, sensor, and disturbance input.

In this diagram:

- $X^{sp}(s)$ is the desired position (set-point),
- $E(s)$ is the error between the desired and measured position,
- $G_c(s)$ is the controller,
- $G_s(s)$ is the plant transfer function,
- $G_m(s)$ is the sensor transfer function,
- $G_d(s)$ represents the disturbance path,
- $X(s)$ is the actual position of the ball.

This representation is introduced at the controller-design stage. The reference input and error signal

are not part of the original plant model, but are defined here to describe how feedback is used to regulate the system output.

8 Open-Loop Stability Analysis

A linear time-invariant system is BIBO stable if all poles of its transfer function have strictly negative real parts.

Using the equilibrium values at $x_{sp} = 0.5$, the coefficients were evaluated numerically in the handwritten derivation. The denominator polynomial becomes

$$s^3 + 15174.90\, s^2 + 243241.11\, s - 45867102.74.$$

The corresponding roots are approximately

$$s_1 \approx -15158.62, \qquad s_2 \approx 47.47, \qquad s_3 \approx -63.75.$$

Since one pole is positive, the open-loop system is not BIBO stable. This is an important conclusion because it shows that the plant cannot regulate itself without feedback. It also explains why controller design is necessary.

9 Control Design

9.1 Controller Choice

A PID controller is selected:

$$G_c(s) = K_p + K_d s + \frac{K_i}{s}.$$

9.2 Controller Selection Justification

Different controller structures were considered before selecting the final design.

A proportional (P) controller was not chosen because it does not eliminate steady-state error. In this system, the presence of disturbances and model uncertainties means that a P controller would result in a non-zero offset from the desired set-point.

A proportional–integral (PI) controller was also considered. While the integral action removes steady-state error, it was determined that the system requires additional damping to reduce oscillations and improve transient response. Based on the system behaviour observed during analysis, the inclusion of a derivative term was necessary.

For these reasons, a proportional–integral–derivative (PID) controller was selected:

$$G_c(s) = K_p + K_d s + \frac{K_i}{s}.$$

The PID controller provides zero steady-state error due to the integral term, while the derivative term improves stability and reduces oscillatory behaviour. Furthermore, analysis of the closed-loop system shows that the proportional gain must satisfy

$$K_p > 447.3.$$

This confirms that the chosen PID structure can satisfy the required performance in terms of stability, accuracy, and transient response.

9.3 Closed-Loop Stability Condition

Using a proportional controller $G_c(s) = K_p$, the closed-loop characteristic equation is obtained from

$$1 + G_c(s)G_s(s) = 0.$$

Since the plant transfer function is

$$G_s(s) = \frac{CF}{s^3 + s^2(-B-E) + sBE - A - CD + AE},$$

the characteristic polynomial becomes

$$s^3 + s^2(-B-E) + sBE - A - CD + AE + K_p CF = 0.$$

Substituting the numerical values obtained from the linearisation gives

$$s^3 + 15174.90\, s^2 + 243241.11\, s + \big(-45867102.74 + 102538.47 K_p\big) = 0.$$

For a third-order polynomial

$$s^3 + a_1 s^2 + a_2 s + a_3,$$

BIBO stability requires the relevant stability conditions to hold. In particular, the constant term must be positive, so

$$-45867102.74 + 102538.47 K_p > 0.$$

Hence,

$$K_p > 447.3.$$

This provides a lower bound on the proportional gain for the closed-loop system and supports the controller design choice.

9.4 Sensor and Disturbance Models

The sensor is modelled as a first-order system:

$$G_m(s) = \frac{K_m}{\tau_m s + 1}.$$

Using $K_m = 1$ and $\tau_m = 0.03$,

$$G_m(s) = \frac{1}{0.03s + 1}.$$

The disturbance path is represented by $G_d(s)$. If the disturbance acts additively at the output, $G_d(s)$ can be taken as unity unless a more detailed disturbance model is introduced.

9.5 Closed-Loop and Load Transfer Functions

From the block diagram, the output can be written as

$$X(s) = G_{CL}(s)X^{sp}(s) + G_L(s)D(s),$$

where the closed-loop transfer function is

$$G_{CL}(s) = \frac{G_c(s)G_s(s)}{1 + G_c(s)G_s(s)G_m(s)}$$

and the load transfer function is

$$G_L(s) = \frac{G_d(s)}{1 + G_c(s)G_s(s)G_m(s)}.$$

These expressions are useful because $G_{CL}(s)$ describes how the system tracks the set-point, while $G_L(s)$ describes how disturbances affect the output.

10 Disturbances and Robustness

Realistic disturbances acting on this system may include:

- friction variation,
- uncertainty in k, b, c, or the inductance model,
- voltage supply variation,
- measurement noise,
- unmodelled rolling losses.

The control system deals with these disturbances through feedback. In particular, the integral action removes steady-state error caused by constant disturbances, while the derivative term can reduce oscillations after sudden changes. A robust design should also be tested under parameter mismatch and against the original nonlinear model.

11 Open-Loop Behaviour

If the system starts at equilibrium and the voltage is increased instantaneously by 1 V in open loop, the current increases and the magnetic force becomes stronger. As a result, the ball is pulled further toward the electromagnet and accelerates down the incline. Because the system is nonlinear, the spring force, damping, and magnetic force all change as the states evolve. Since the plant is open-loop unstable, feedback is required to regulate the motion and return the system to the desired operating point.

12 Simulation Results and Discussion

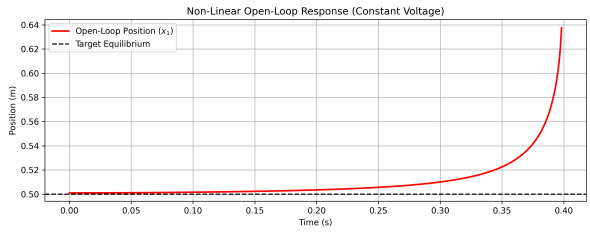

Figure 2: Open-loop nonlinear response to a +1mm initial position disturbance

This graph demonstrates the open-loop instability of the magnetic suspension system at the $x_{sp} = 0.50m$ operating point. With the PID controller disabled and a constant equilibrium voltage applied, a $+1mm$ positional disturbance causes the mass to rapidly and unstably diverge toward the electromagnet. This confirms that the uncontrolled plant is fundamentally unstable, validating the absolute necessity of closed-loop feedback control.

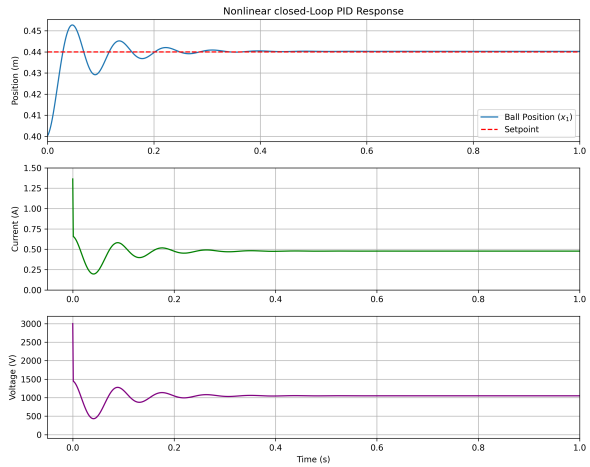

Figure 3: Closed-loop Non-linear response

This graph illustrates the closed-loop tracking performance of the simplified 3-state nonlinear model. The ball is initially displaced to $x_1 = 0.40m$ and commanded to reach the $x_{sp} = 0.44m$ setpoint. By removing the sensor dynamics (τ_m) used in previous tests, the PID controller operates with an ideal, zero-latency measurement. Consequently, the position response is exceptionally smooth, achieving a settling time of approximately 0.4 seconds with negligible overshoot and zero steady-state error. The actuator demand plot (bottom) confirms the controller initially rises to the $3000V$ hardware limit to rapidly accelerate the mass, before settling precisely at the theoretically calculated equilibrium voltage ($V_{eq} \approx 1043V$).

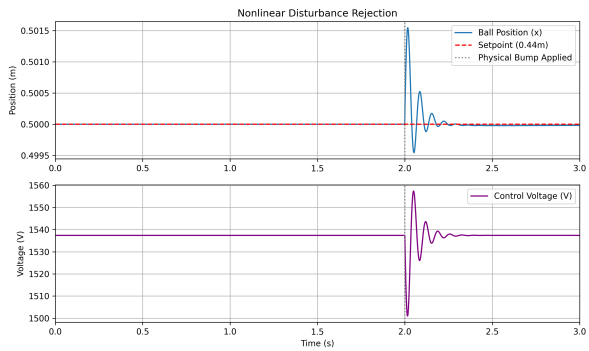

Figure 4: Closed-loop Non-linear disturbance response

This graph demonstrates the closed-loop disturbance rejection of the non-linear plant at the $x_{sp} = 0.44m$ equilibrium, incorporating the $\tau_m = 0.03s$ sensor. The system is hovering stably when a sudden physical step-disturbance (a $+0.15\ m/s$ velocity impulse) is injected at $t = 2.0s$. Unlike an idealized zero-latency model, the inclusion of the realistic sensor delay prevents instantaneous controller reaction. This phase lag results in a visible transient overshoot and minor oscillation before the PID controller accumulates the necessary integral effort to successfully restore the mass to the target setpoint. The actuator demand plot confirms the rapid, realistic voltage fluctuations required to arrest the sudden velocity spike without saturating the power supply.

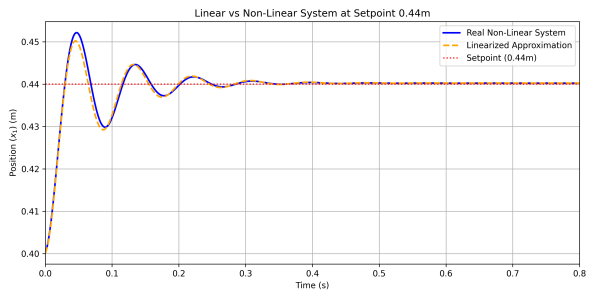

Figure 5: Nonlinear vs Linear response at $x_{sp} = 0,44$

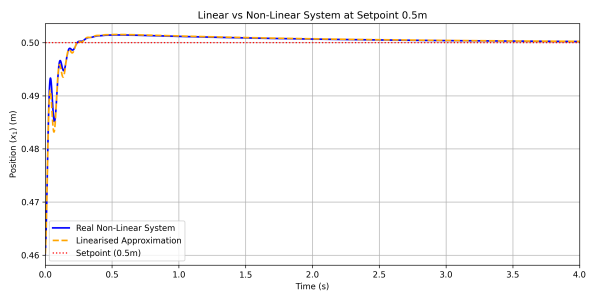

Figure 6: Nonlinear vs Linear response at $x_{sp} = 0,5$

Comparing these two responses clearly illustrates the limitations of small-signal linearisation in a highly non-linear system:At $x_{sp} = 0.44m$ (Further from magnet): The linearised model accurately tracks the true non-linear plant. The magnetic force curve is relatively gradual here, allowing the fixed PID controller to stably manage the $-4cm$ displacement with minimal oscillation.At $x_{sp} = 0.50m$: The linear approximation completely breaks down. Because the $1/y^2$ magnetic force is exponentially steeper at this distance, the $-4cm$ step pushes the system far outside its valid linear region. Both models settle at 0.5m but take much longer (around the 4 second mark).

13 Conclusion

A nonlinear model of the wooden-ball system was derived from mechanical and electrical principles and written in state-space form. The system was linearised around the equilibrium point $x_{sp} = 0.5\,\mathrm{m}$, and the Laplace transform was used to obtain a transfer function for open-loop analysis. The resulting transfer function shows that the open-loop system is unstable, since one pole lies in the right-half plane. This justifies the use of feedback control. A PID controller was selected as a practical and effective choice for achieving stability, zero steady-state error, and improved damping. Although the local linear model is useful for controller design, final validation should always be performed using the nonlinear plant and realistic disturbances.

A Interview Preparation Notes

What are the states, input, and output? The simplified states are position, velocity, and current. The input is voltage, and the output in the simplified state-space model is position.

Why are the sensor dynamics not included in the state-space model above? Because the simplified plant model is written using the mechanical and electrical states only. The sensor is then introduced later in the controller-design stage through the transfer function $G_m(s)$.

How accurate is the linearisation? It is accurate only near the operating point. For larger deviations, the nonlinear magnetic-force and inductance terms become more significant.

Why derive the transfer function? The transfer function makes stability analysis and controller design easier by representing the input-output behaviour in the Laplace domain.

Is the open-loop system BIBO stable? No. One pole of the denominator is positive, so the open-loop system is unstable.

Why use PID? PID provides zero steady-state

error and improved damping. A P controller would leave an offset, and a PI controller was considered insufficient because the derivative term is needed to improve transient behaviour.

B Collaboration Report

Replace this appendix with your real collaboration evidence before submission.

Collaboration tools: GitHub, Overleaf, MS Teams.

Project management approach: weekly meetings, clear task allocation, and version control.

Individual contributions: each team member should state their exact technical contribution.

Challenges and solutions: explain how modelling disagreements, scheduling problems, or coding issues were resolved.